

AIM:

To understand the **lifecycle of an Activity and Fragment** in Android and observe how different lifecycle methods are invoked during various states of the application.

What is Lifecycle?

- The lifecycle refers to the different states an Activity or Fragment goes through from creation to destruction.
- Android automatically calls lifecycle methods based on user actions, such as:
 - Opening the app
 - Switching apps
 - Rotating the device
 - Closing the app

Why Lifecycle Management is Important

- To manage resources efficiently
- To save and restore user data
- To avoid memory leaks
- To ensure smooth app behavior during background/foreground transitions

Activity Lifecycle Methods

Method	Purpose
onCreate()	Activity is created
onStart()	Activity becomes visible
onResume()	Activity starts interacting with user
onPause()	Activity partially hidden
onStop()	Activity no longer visible
onDestroy()	Activity is destroyed

Fragment Lifecycle (Brief)

- Fragment lifecycle is similar to Activity but depends on the host Activity.
- Important methods include:
 - onCreateView()
 - onStart()
 - onResume()
 - onPause()
 - onStop()
 - onDestroyView()

How Lifecycle is Observed

- Lifecycle events are observed using:
 - Logcat messages
 - TextView updates

Procedure

1. Create an Android project using Empty Activity.
2. Open MainActivity.kt.
3. Override Activity lifecycle methods.
4. Add Log statements in each lifecycle method.
5. Run the app.
6. Perform actions:
 - Open app
 - Press Home button
 - Reopen app
 - Rotate screen
 - Close app
7. Observe lifecycle callbacks in Logcat.

STEP 1: Create New Project

1. Open Android Studio
2. Click New Project
3. Select Empty Views Activity
4. Click Next
5. Enter:
 - Name: LifecycleDemo
 - Language: Kotlin
 - Minimum SDK: API 21
6. Click Finish
7. Wait until Build Successful

STEP 2: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <FrameLayout

        android:id="@+id/fragment_container"

        android:layout_width="match_parent"

        android:layout_height="0dp"

        android:layout_weight="1"/>

    <TextView

        android:id="@+id/tvLifecycle"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Lifecycle Output:"

        android:textSize="16sp"

        android:padding="10dp"/>

</LinearLayout>
```

STEP 3: Edit MainActivity.kt

```
package com.example.lifecycledemo

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var tvLifecycle: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        tvLifecycle = findViewById(R.id.tvLifecycle)
        tvLifecycle.append("\nActivity onCreate")

        supportFragmentManager.beginTransaction()
            .replace(R.id.fragment_container, DemoFragment())
            .commit()
    }

    override fun onStart() {
        super.onStart()
        tvLifecycle.append("\nActivity onStart")
    }
}
```

```
override fun onResume() {  
    super.onResume()  
    tvLifecycle.append("\nActivity onResume")  
}
```

```
override fun onPause() {  
    super.onPause()  
    tvLifecycle.append("\nActivity onPause")  
}
```

```
override fun onStop() {  
    super.onStop()  
    tvLifecycle.append("\nActivity onStop")  
}
```

```
override fun onDestroy() {  
    super.onDestroy()  
    tvLifecycle.append("\nActivity onDestroy")  
}  
}
```

STEP 4: Create Fragment Class

1. Right-click package name
2. Select New → Kotlin Class/File
3. Name: DemoFragment
4. Type: Class
5. Click OK

STEP 5: Write DemoFragment.kt Code

```
package com.example.lifecycledemo

import android.os.Bundle
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import androidx.fragment.app.Fragment

class DemoFragment : Fragment() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        (activity as MainActivity).tvLifecycle.append("\nFragment onCreate")
    }

    override fun onCreateView(
        inflater: LayoutInflater,
        container: ViewGroup?,
        savedInstanceState: Bundle?
    ): View? {
        (activity as MainActivity).tvLifecycle.append("\nFragment onCreateView")
        return inflater.inflate(R.layout.fragment_demo, container, false)
    }

    override fun onResume() {
        super.onResume()
        (activity as MainActivity).tvLifecycle.append("\nFragment onResume")
    }
}
```

```

override fun onPause() {

    super.onPause()

    (activity as MainActivity).tvLifecycle.append("\nFragment onPause")

}

override fun onDestroyView() {

    super.onDestroyView()

    (activity as MainActivity).tvLifecycle.append("\nFragment onDestroyView")

}

override fun onDetach() {

    super.onDetach()

    (activity as MainActivity).tvLifecycle.append("\nFragment onDetach")

}

}

```

STEP 6: Create Fragment Layout

res → layout → New → Layout Resource File

- File name: fragment_demo
- Root element: TextView

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<TextView
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:text="Fragment Lifecycle Demo"
```

```
    android:textSize="22sp"
```

```
    android:gravity="center"/>
```


STEP 7: Run Using Device Manager

- Click Device Manager
- Start an Emulator
- Click Run ►
- App opens on device

Output

